RAHUL MITRA

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EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

WORK EXPERIENCE

Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone) **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine ingesting each site's shelf coordinates, dimensions, and operating rules, letting one model adapt to any STMicroelectronics warehouse floor plan without per-site rework.

Global Data Engineer (Operational AI Systems), Part-Time, Abbott Laboratories (Contract) **Jun 2026 – Present**

- Developed ADAM, an offline tool-calling assistant on a locally hosted Gemma 4 E4B model (llama.cpp) answering spray-dryer setpoint, process, and signal questions for 6 dryers across 4 global sites, giving engineers self-serve answers to decisions previously requiring APC specialists, with rollout to Abbott servers scheduled.
- Built a Seeq signal-analytics toolkit (Kalman smoothing, Welch spectra, lag and coherence, PELT change points, matrix-profile motifs and discords, an 18-feature fingerprint) and a 5-stage offline handbook RAG pipeline: GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain retrieval.

Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract) **Jan 2026 – Jun 2026**

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Validated a non-destructive, 15-stage changeover-data pipeline against 5 years of unseen raw data with zero execution failures across operator- and machine-generated anomalies.

Software Engineer (Full Stack & Cloud), People's Association (Pro bono) **Sep 2025 – Sep 2026**

- Recruited into PA's Sparks Citizens Developer network as a community innovator in recognition of prior public-good initiatives, then designed, built, and deployed Churp, a resident-driven garden-plot balloting platform now live in Tengah and slated for national rollout, seeded by a S$20K Sparks Community Innovation Fund award.
- Owned most software development and deployment across admin and public Next.js frontends, a NestJS backend, and GIS mapping.
- Integrated Singpass Login (OIDC) to secure resident sign-in and Myinfo to verify identity and address, keeping garden-plot balloting fair for 150 residents signed up at release.
- Led deployment planning, multi-repo coordination, and stakeholder management in an 11-person team across People's Association, Tengah community partners, and PA ICT (GovTech).

RESEARCH EXPERIENCE

Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge) **Jul 2026 – Oct 2026**

- Built vision systems turning raw camera feeds into perception-ready frames across 3 workstreams: camera ISP, super-resolution, and image restoration under real-world blur.

PROJECTS

Team Lead & Finalist, Swarmline: SG Defence Tech Hackathon (1,300+ applicants) **Sep 2026**

- Implemented a deterministic Unreal Engine 5.8 C++ simulation with Claude Code; under a ground-link jam, 30 drones confirmed all 8 walking targets in 10 of 10 seeded runs, versus 2.1 of 8 on average for an operator-run fleet.

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams) **Jan 2026**

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once.

CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

Independent Bug Bounty Researcher, YesWeHack **May 2024 – Present**

- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and access-control weaknesses.

Elderly Care Volunteer, NTUC Health Active Ageing Centre **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check on residents' wellbeing, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

SKILLS AND CERTIFICATIONS

Languages: Python, TypeScript, JavaScript, Java, Rust, C++, SQL, Bash, Excel VBA
Web and Cloud: React, Next.js, Three.js, NestJS, FastAPI, Flask, Docker, AWS (ECS Fargate, RDS, ALB, WAFv2, Lambda, API Gateway, Route 53, ACM, KMS, Secrets Manager, IAM OIDC), Terraform, AWS SAM, Azure App Service, GitHub Actions and Bitbucket CI/CD, Supabase, Singpass Login and Myinfo (OIDC), API design, GIS mapping
AI and Machine Learning: RAG, AI agents, schema-constrained tool calling, local LLM serving (llama.cpp, Gemma), vector search, LangChain, PyTorch, ONNX, Chroma, BERT/transformer fine-tuning, model evaluation, deep RL (PPO, A2C, DDPG, DQN), GPU training (CUDA) on ASPIRE 2A HPC, Seeq time-series analysis (Kalman filtering, spectral analysis, change-point detection, matrix profile)
Security: OWASP Top 10, Burp Suite, Wireshark, OAuth2/JWT, SSRF/CSRF/injection testing, fuzzing, secure design
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery